



Programming Foundations Level 1



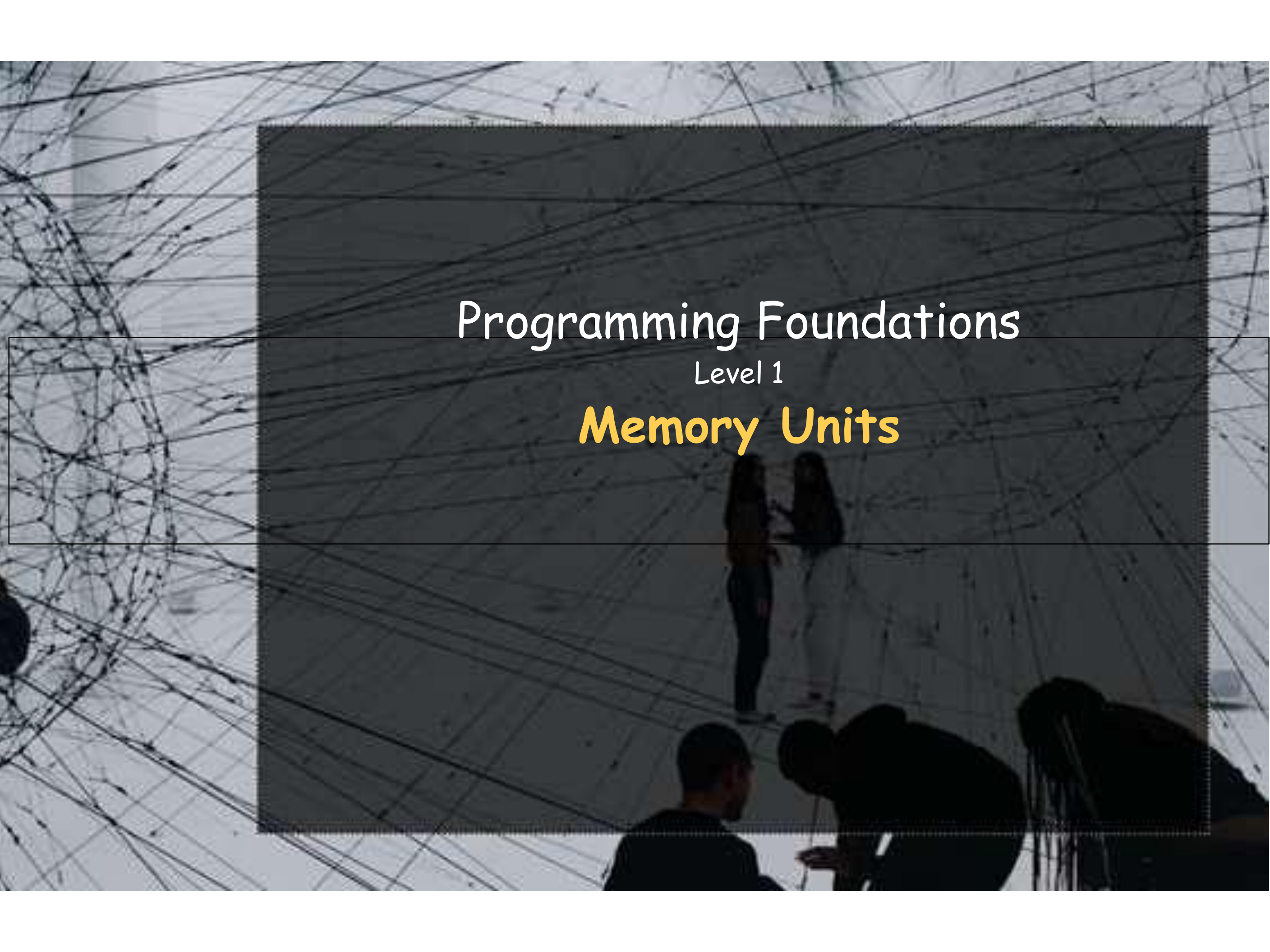
Programming Foundations

Level 1

Computer Speeds and Units

Computer Speeds & Units (Bits, Bytes, Hz, and Architectures)

Understanding how we measure data and speed is foundational: from bits & bytes to kilobytes through yottabytes, from CPU cycles (Hz) to how quickly the CPU reaches memory, and why 32-bit vs 64-bit still matters.

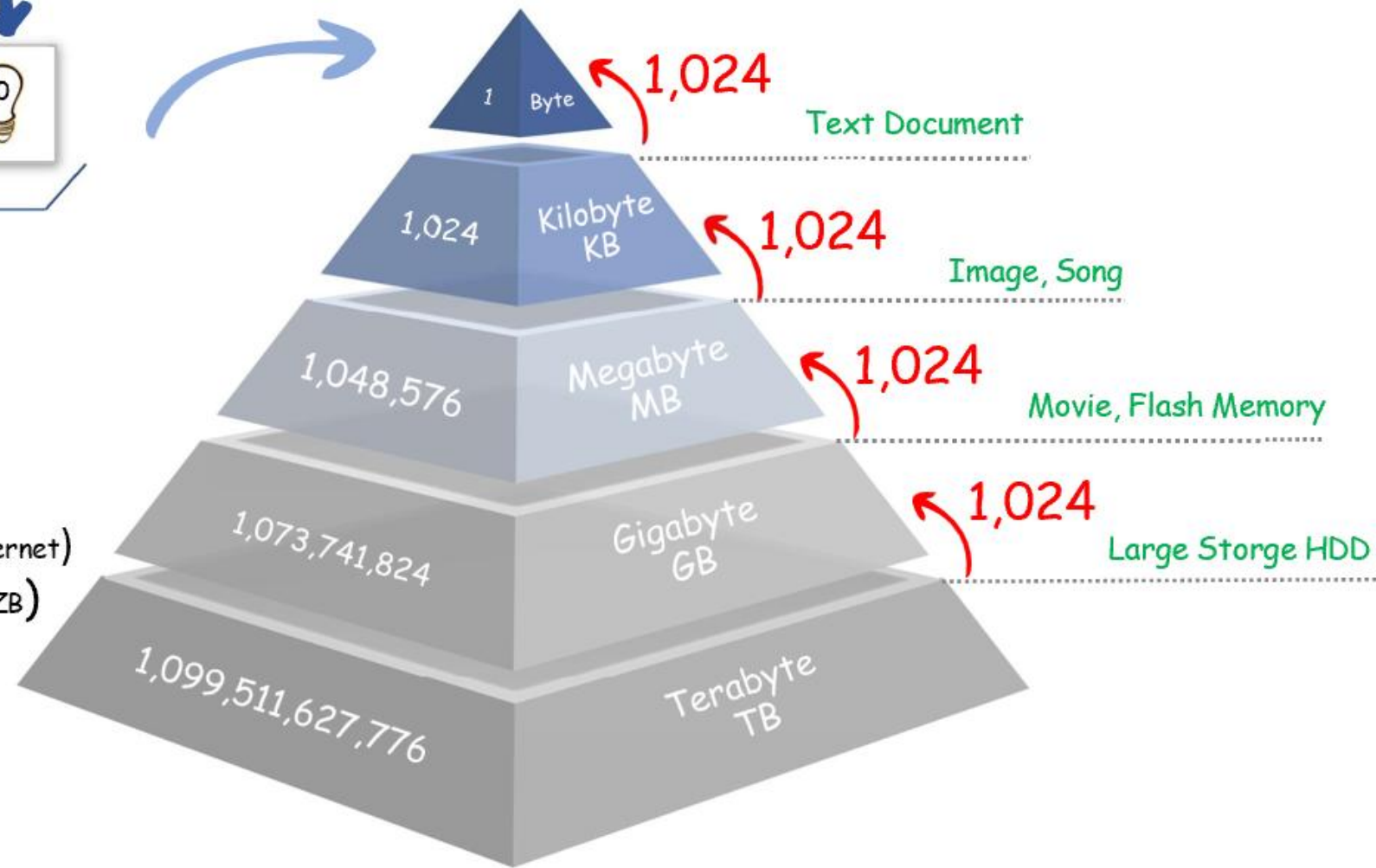
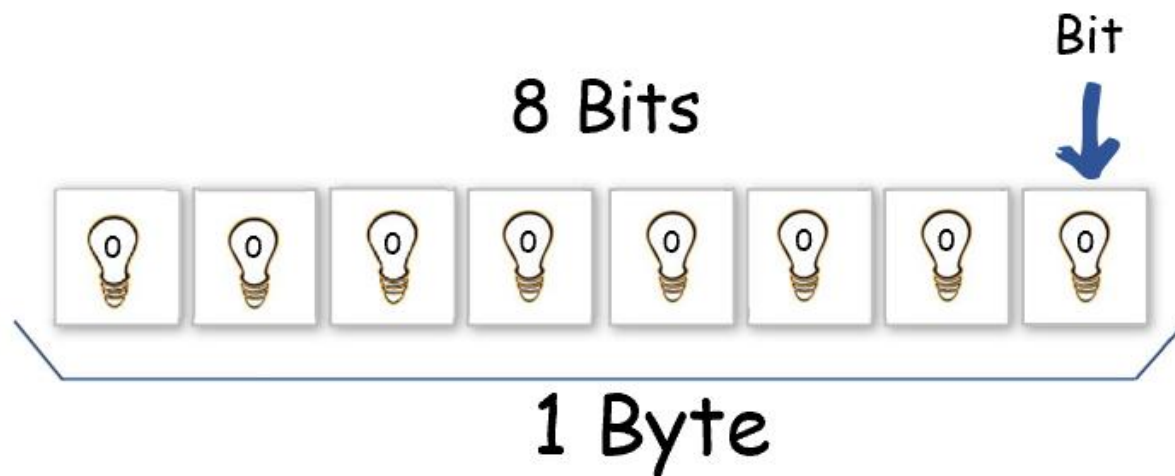
The background of the slide is a dark, abstract image. It features a dense, intricate network of thin, light-colored lines that resemble a complex web or a neural network. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are visible in the foreground, some appearing to be looking at a screen or device. The overall aesthetic is technical and modern.

Programming Foundations

Level 1

Memory Units

Memory Units...

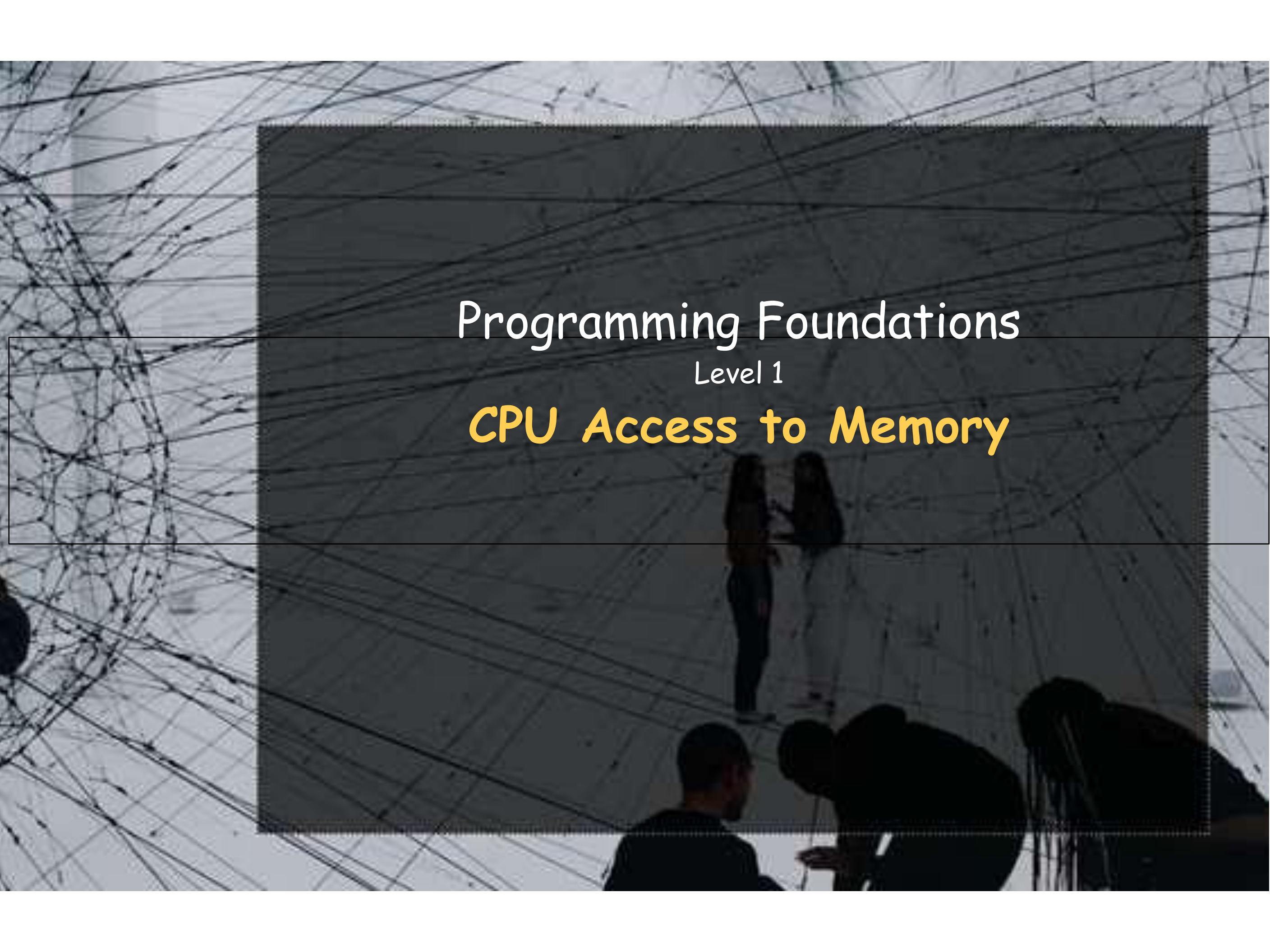


There are more:

- Petabyte (PB) 10^{15} (Facebook Servers 10+ PBs)
- Exabyte (EB) 10^{18} (amount of data transfer over the internet)
- Zettabyte (ZB) 10^{21} (whole data in the world just a few ZB)
- Yottabyte (YB) 10^{24} (Huge, No practical use)

Memory Units...

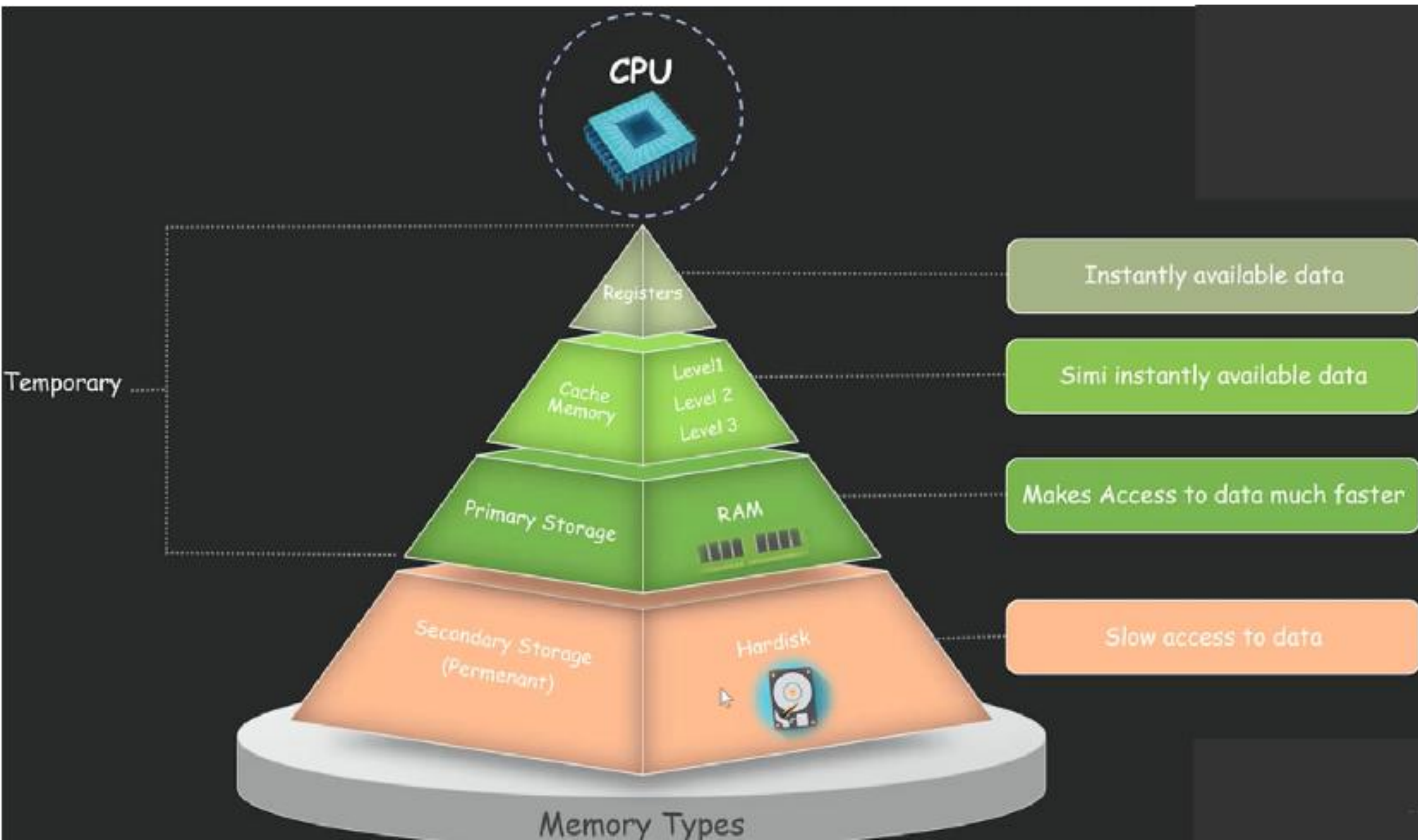
- Bit = the smallest unit (0 or 1).
- Byte = 8 bits.
- Binary-scaled units (base 2) commonly taught to beginners:
 - 1 KB = 1,024 bytes
 - 1 MB = 1,024 KB
 - 1 GB = 1,024 MB
 - 1 TB = 1,024 GB
- Even Larger : PB, EB, ZB, YB with real-world anchors (e.g., Facebook servers in PBs, global internet traffic in EBs, "whole data in the world" a few ZBs).
- Note for learners: Operating systems and storage vendors sometimes use decimal (10^3) labels; this lesson uses the binary ($2^{10}=1,024$) view shown in the slides.



Programming Foundations

Level 1

CPU Access to Memory



Memory Units...

The closer the storage is to the CPU, the faster the access:

- Registers (inside CPU): instantly available values for current instructions.
 - Cache (on-chip): very fast access to recently used data/instructions.
 - Main Memory (RAM): fast, but slower than cache.
 - Secondary Storage (SSD/HDD): slowest among these tiers, used for long-term storage.
- Slides emphasize: "Instantly / semi-instantly available data... makes access to data much faster."

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in motion, which adds a sense of activity and collaboration to the scene.

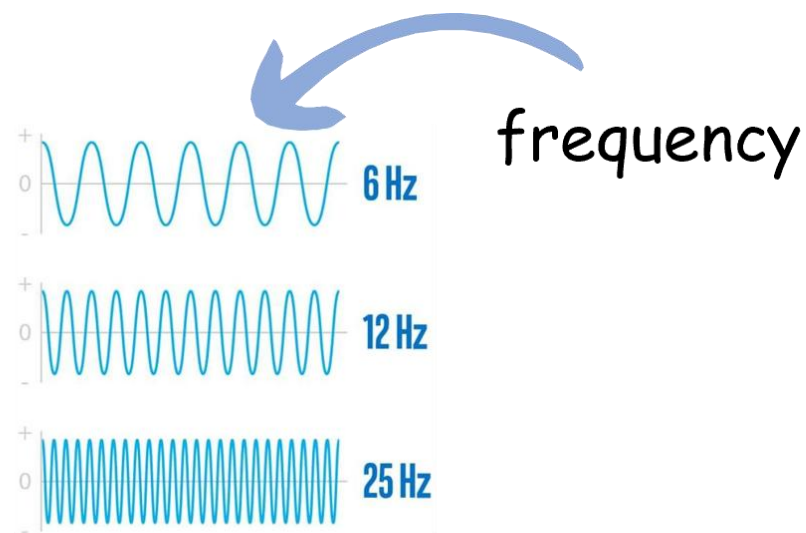
Programming Foundations

Level 1

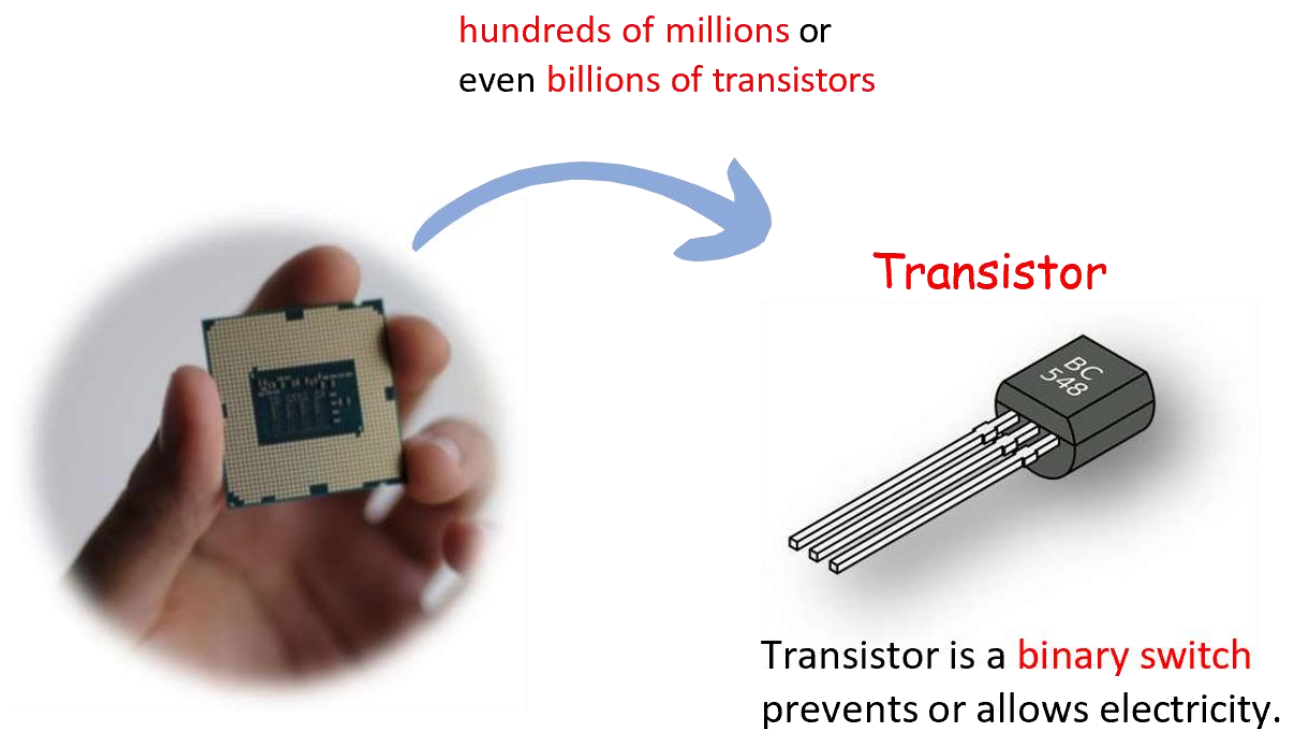
CPU Speed

CPU Speed

Higher clock speed means a faster CPU (within the same generation)
Clock speed (also "clock rate" or "frequency") is one of the most significant.
A CPU with a clock speed of 3.2 GHz executes 3.2 billion cycles per second.



Hertz HZ → Cycles per second
(Clock Cycle)



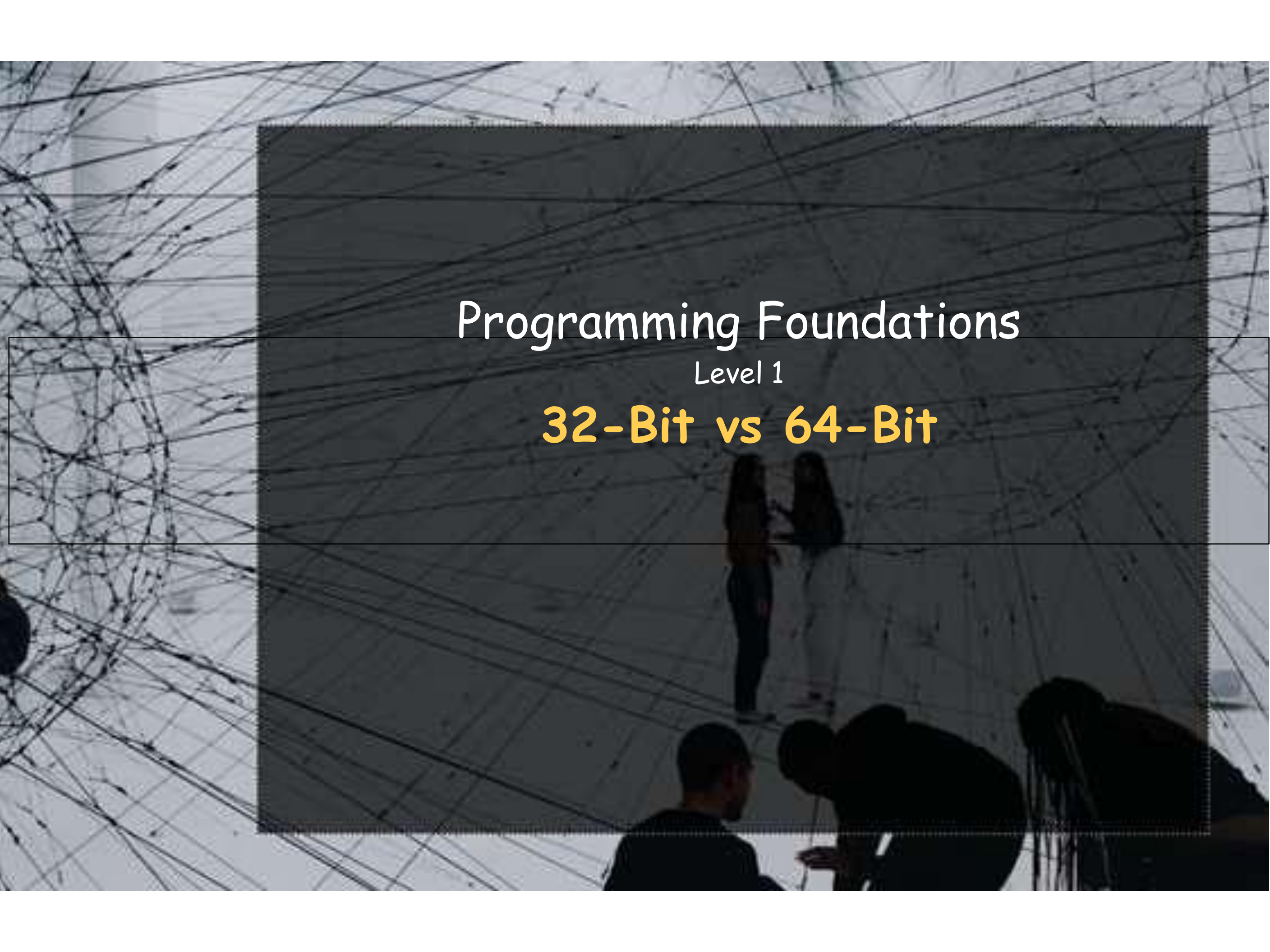
During each cycle, billions of transistors within the processor open and close.

Memory Units...

- Clock speed / frequency measures how many cycles per second the CPU performs, in Hertz (Hz).
- 3.2 GHz = 3.2 billion cycles per second.
- On each cycle, billions of transistors open/close (binary switching) to execute instructions.
- Transistor = binary switch that prevents or allows electricity (OFF/ON → 0/1).

Device specifications

Device name	MSaqer-Mac
Processor	Intel(R) Core(TM) i9-9880H CPU @ 2.30GHz 2.30 GHz
Installed RAM	16.0 GB (15.9 GB usable)
Device ID	DA94F72E-CA16-4D3D-9AFE-78DA0D409BA6
Product ID	00329-10286-19105-AA841
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display



Programming Foundations

Level 1

32-Bit vs 64-Bit

Device specifications

Device name	MSaqer-Mac
Processor	Intel(R) Core(TM) i9-9880H CPU @ 2.30GHz 2.30 GHz
Installed RAM	16.0 GB (15.9 GB usable)
Device ID	DA94F72E-CA16-4D3D-9AFE-78DA0D409BA6
Product ID	00329-10286-19105-AA841
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

32-Bit

VS

64-Bit

Less processing power, older, slower, and less secure

More processing power, newer, faster, and more secure

2^{32} (or 4,294,967,296) bytes

2^{64} or 18,446,744,073,709,551,616) bytes

Can allocate only 4GB of memory (RAM

Can allocate much more RAM

Referred to as x86 or x86-32

Referred to as x64 or x86-64

Can only run a 32-bit operating system software and 32-bit

Can run both 64-bit and 32-bit operating systems and software

Differences

Note: if you have installed 32-bit operating system on a 64-bit computer, then it can run 32-bit software only.

Address space & capacity

- $2^{32} = 4,294,967,296$ bytes ≈ 4 GB addressable by a 32-bit system.
- $2^{64} = 18,446,744,073,709,551,616$ bytes ≈ 16 EB (vastly larger) for 64-bit.

Naming

- 32-bit: x86 / x86-32
- 64-bit: x64 / x86-64

Performance

- 64-bit can handle **twice as many bits per cycle** (within the same generation), enables larger data types and more RAM.

Compatibility (as shown in slides)

- 32-bit CPU $\rightarrow$ runs 32-bit OS & 32-bit software only.
- 64-bit CPU $\rightarrow$ can run 64-bit & 32-bit OS/software.
- If you install a 32-bit OS on a 64-bit CPU, you can run 32-bit software only (you won't benefit from 64-bit apps or large RAM).

RAM reality

- 32-bit OS: can allocate only about 4 GB.
- 64-bit OS: can allocate **much more** (practically limited by hardware/edition).
If you have ≥ 4 GB RAM, you should use 64-bit CPU + 64-bit OS to utilize it fully.

How to Know Your Device Specifications on Windows

① Basic Specifications (Processor, RAM, and System Type)

Method 1: Using "About This PC"

① Press **Windows + I** → open **Settings**.

② Go to **System** → **About**.

③ Under **Device specifications**, you'll see:

 **Processor:** CPU model and speed (e.g., *Intel Core i7-10750H @ 2.60 GHz*).

 **Installed RAM:** total physical memory (e.g., *16 GB*).

 **System type:** whether your Windows and processor are **32-bit** or **64-bit**.

→ *Example: "64-bit operating system, x64-based processor."*



2 GPU & Graphics Card Details



Method 2: Using Device Manager

- 1 Press **Windows + X** → select **Device Manager**.
- 2 Expand **Display Adapters**.
- 3 You'll see one or more GPUs listed:
Integrated GPU (e.g., *Intel UHD Graphics*).
Dedicated GPU (e.g., *NVIDIA GeForce RTX 3060* or *AMD Radeon 6600*).

RX



Alternative (fastest)

Press **Windows + R**, type **dxdiag**, press **Enter**.

Go to the **Display** tab → see GPU name, driver version, and VRAM (memory).

3 Storage (SSD / HDD Type & Capacity)

Method 3: Using File Explorer

- 1** Open **This PC**.
- 2** Right-click any drive → **Properties**.
- 3** View:

Used space / Free space / Total capacity.

To know the physical drive type

Press **Windows + S**, type **"Defragment and Optimize Drives"** → open.

Under **Media type**, you'll see:

"Solid State Drive (SSD)" or

"Hard Disk Drive (HDD)".



4 Advanced Full System Information



Method 4: Using System Information Tool

- 1 Press **Windows + R**, type msinfo32, press **Enter**.
- 2 This opens the **System Information** window showing:
 - OS version
 - Manufacturer, model
 - BIOS version
 - Installed RAM
 - Processor details
 - System type (32/64-bit)
 - Boot mode, etc.



You can also **Export** this report:

File → Export → choose location.

5 Using Command Prompt or PowerShell (for advanced users)



Command Prompt:

Press **Windows + R**, type **cmd**, press **Enter** then use the following commands:

CPU info → `wmic cpu get name`

RAM info → `wmic computersystem get totalphysicalmemory`

System type → `wmic os get osarchitecture`



PowerShell:

Run → `Get-ComputerInfo | Select CsName, OsName, OsArchitecture, CsSystemType, CsTotalPhysicalMemory`



Activity



Instructions:

- 1 Convert each: **8 KB**, **2 MB**, **3 GB** into bytes (use **1 KB = 1,024 bytes**).
- 2 Order these by **fastest access**: **RAM**, **Registers**, **SSD**, **L1 Cache**. Explain why.
- 3 Your laptop has **8 GB RAM** and a **64-bit CPU**, but a **32-bit OS** installed. List **two limitations** you'll face.
- 4 A CPU runs at **3.6 GHz**. How many cycles per second is that? What happens during each cycle (at a high level)?



Activity Solution



Solution:

- 1 Convert each: 8 KB, 2 MB, 3 GB into bytes (use 1 KB = 1,024 bytes).
 $8 \text{ KB} = 8 \times 1,024 = 8,192 \text{ bytes}$
 $2 \text{ MB} = 2 \times 1,024 \times 1,024 = 2,097,152 \text{ bytes}$
 $3 \text{ GB} = 3 \times 1,024^3 = 3,221,225,472 \text{ bytes}$
- 2 Order these by fastest access: RAM, Registers, SSD, L1 Cache. Explain why.
Registers → **L1 Cache** → **RAM** → **SSD** (closest to farthest from CPU → fastest to slowest).
- 3 Your laptop has 8 GB RAM and a 64-bit CPU, but a 32-bit OS installed. List two limitations you'll face.
You can't fully use > 4 GB RAM; you can run 32-bit apps only (no 64-bit software benefits)
- 4 A CPU runs at 3.6 GHz. How many cycles per second is that? What happens during each cycle (at a high level)?
3.6 GHz = 3.6 billion cycles/sec; during each cycle, the CPU's transistors switch, moving/processing data per the current instruction.

Conclusion

- ✓ **Bits & Bytes** define data size; **KB**→**MB**→**GB**→**TB** (1,024 steps) scale storage and memory.
- ✓ **CPU**↔**Memory** speed hinges on **distance**: Registers/Cache (closest, fastest) → RAM → SSD/HDD.
- ✓ **Hz** = cycles/sec; more cycles (same gen) → more potential work; **transistors** toggle every cycle.
- ✓ **64-bit** architectures unlock **far larger address space** and **broader capability** than **32-bit**—and your OS choice must match your hardware to realize the benefits.

Thank you 😊